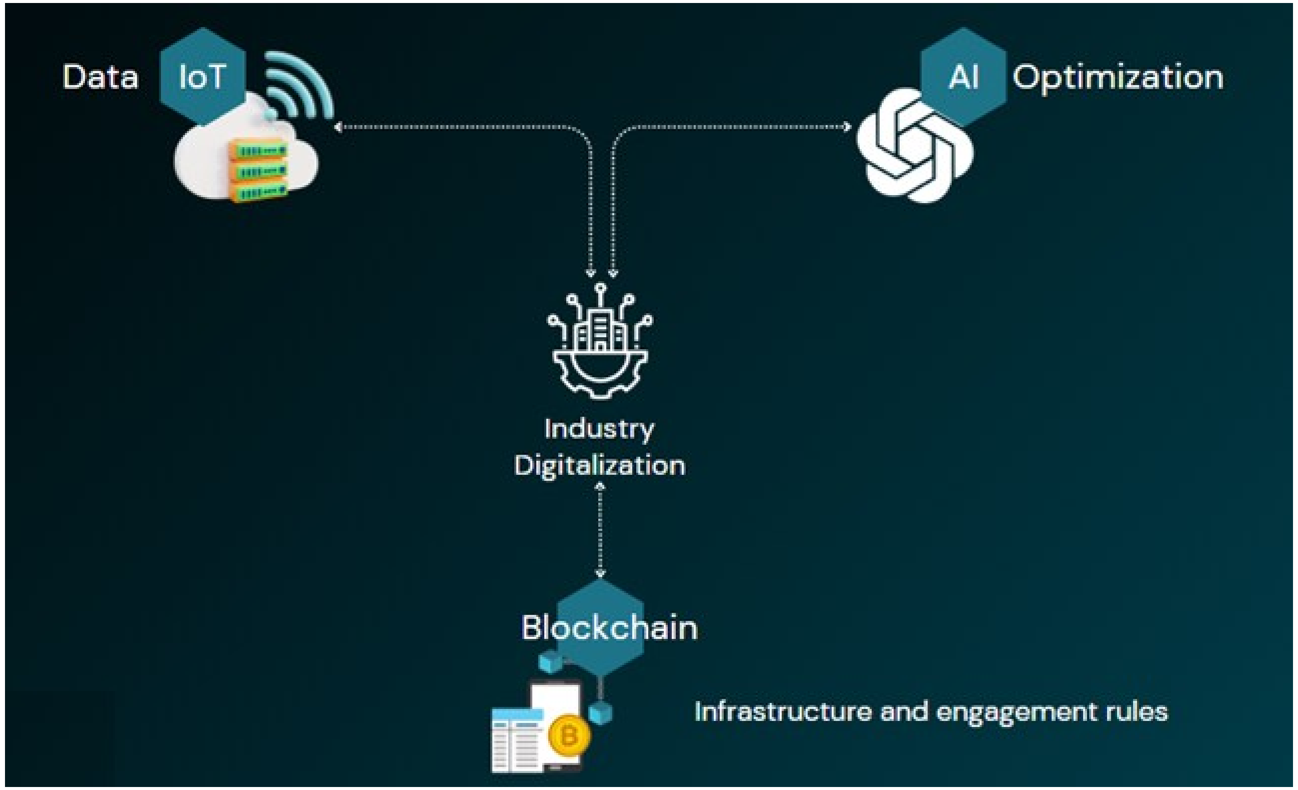
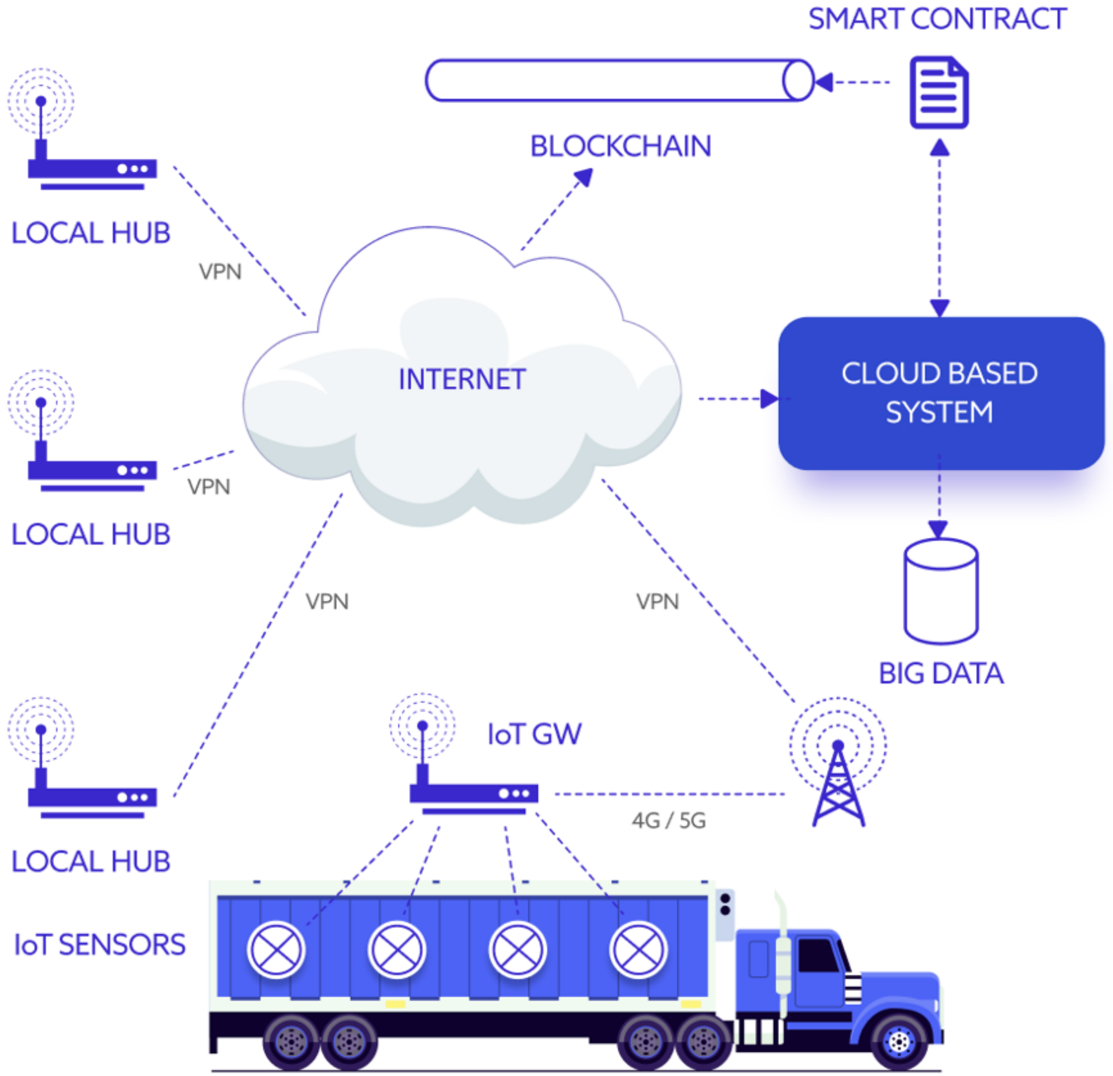
Modern Asset Tracking with IoT, AI & Blockchain



How IoT, AI & Blockchain is transforming Supply Chain

Moden Asset Tracking with IoT, AI and Blockchain



Changing paradigm in Global Supply Chain
From Visibility to Intelligence to Trust

- **GLOBAL, DISTRIBUTED SUPPLY CHAINS**
- **HIGH-VALUE AND SENSITIVE ASSETS**
- **NEED FOR REAL-TIME VISIBILITY**
- **COMPLIANCE, SECURITY, AND COST PRESSURE**

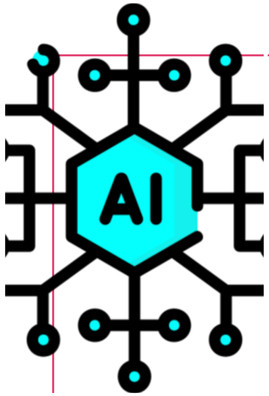
Evolution of Asset Tracking

- Manual logs & barcode systems
- GPS & RFID-based tracking
- IoT-enabled real-time monitoring
- AI-driven intelligence
- Blockchain-enabled trust

Role of IOT & AI in Asset Tracking

Role of IoT in Asset Tracking

- Sensors, devices, and gateways
- Location, temperature, motion, health
- Real-time data generation
- Edge & cloud connectivity
- GPS & RFID-based tracking



What AI Adds to Asset Tracking (AIoT)

- Predictive analytics
- Anomaly detection
- Demand and route optimization
- Automated decision-making

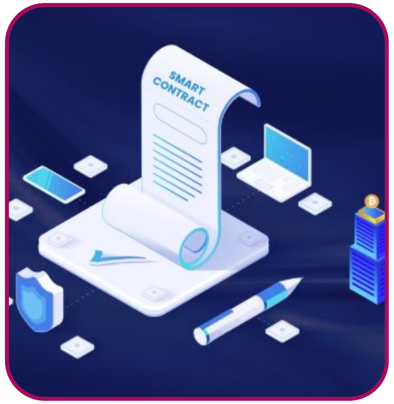
Limitations of IoT + AI Alone

- Data tampering risks
- Centralized control
- Trust issues across partners
- Audit & compliance challenges

Why Blockchain for Asset Tracking

Blockchain for Asset Tracking

- Immutable ledger
- Decentralized trust
- Shared single source of truth
- Tamper-proof records



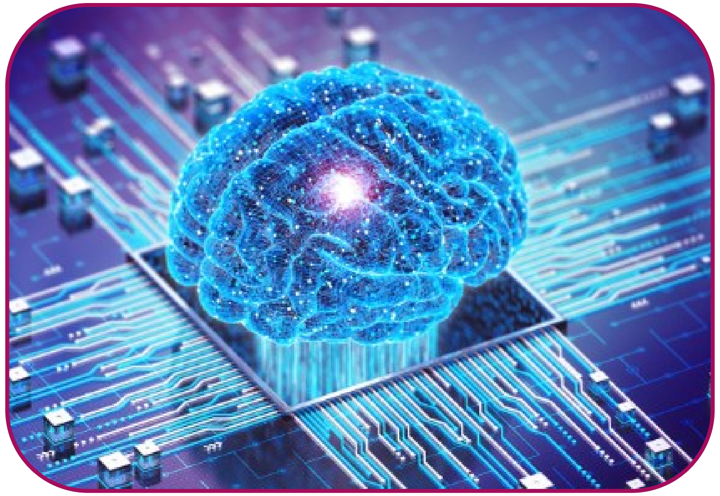
Smart Contracts in Asset Tracking

- Automated rules & workflows
- Condition-based triggers
- Payments, alerts, compliance
- Reduced manual intervention

Integrated Architecture: IoT + AI + Blockchain

Together, they redefine asset tracking & End-to-end asset lifecycle control

IoT → Data generation → Enables visibility



AI → Intelligence & prediction → enables intelligence

Blockchain → Trust & automation → enables trust & automation

Integrated Architecture: IoT + AI + Blockchain

Business Benefits for Organizations adopting IoT, AI, and blockchain

- Real time tracking & monitoring
- Predict delays and failures
- Ensure trusted records
- Reduced losses & theft
- Improved operational efficiency
- Faster dispute resolution
- Regulatory compliance
- Higher customer trust

Key Challenges & Considerations

- Device security
- Data quality
- Scalability
- Integration with legacy systems
- Cost vs ROI

What is Blockchain

Blockchain — is foundation of decentralized systems where no single authority controls the data. This idea has given rise to decentralized economies,

- **Decentralized Economies**
- **Decentralized Autonomous Organizations (DAO)**
- **Web 3**
- **Cryptocurrencies**
- **D-Apps**
- **Decentralized Finance (De-Fi)**
- **Initial Coin Offering (ICO)**
- **Digital Tokens**
- **Non-Fungible Tokens (NFT)**
- **Smart Contracts**
- **Solidity**

What is Blockchain

Blockchain Development : Basics to Advanced

Understand Blockchain, Ethereum & Smart Contracts with Hands-On Coding

https://tinyurl.com/BConUdemy


Decentralized Economies

Decentralized Autonomous Organizations

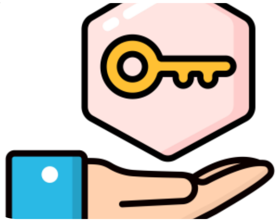
Web 3


Cryptocurrenci
es


D-Apps


Decentralized Finance (De-Fi)

Initial Coin Offering (ICO)


Digital Tokens

Non-Fungible Tokens (NFT)

Smart Contracts


Solidity